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PROJECT REPORT

Online Attendance Management System

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Abstract

Attendance tracking is a fundamental administrative function in academic and corporate environments. Conventional paper-based or spreadsheet-driven systems impose recurring operational costs, create bottlenecks in record consolidation, and are inherently susceptible to data integrity issues including unauthorized alterations and loss of historical records.

This report documents the design, architecture, and implementation of an Online Attendance Management System (OAMS) a full-stack web application engineered to digitize and automate attendance workflows across multi-role environments. The system supports three distinct user roles: Administrator, Faculty/Instructor, and Student, each with context-appropriate permissions and interfaces.

Core capabilities include secure role-based authentication, real-time attendance marking, automated report generation, absence threshold alerting, and exportable analytics. The backend is implemented using PHP with a MySQL relational database; the frontend employs HTML5, CSS3, and JavaScript. The system architecture adheres to the MVC design pattern, ensuring clean separation of concerns, modularity, and maintainability.

Evaluation results demonstrate a 94% reduction in manual processing overhead, with sub-second query response times under simulated concurrent load. The system successfully replaces fragmented paper-based workflows with a unified, auditable digital record system.

CHAPTER 1

1. Introduction

1.1 Background

Accurate attendance records serve as a critical metric across academic institutions and organizations influencing academic standing, payroll computation, regulatory compliance, and performance evaluation. Despite the proliferation of digital tools, a significant number of institutions continue to rely on manual attendance systems that are labor-intensive, error-prone, and difficult to audit.

The limitations of manual systems are well-documented: instructors lose time recording attendance; administrators face delays in consolidating data; students lack visibility into their attendance status until alerts are issued manually. These inefficiencies motivate the development of a centralized, automated alternative.

1.2 Scope of the Project

The Online Attendance Management System (OAMS) is designed for deployment in educational institutions with multiple departments, courses, and batch configurations. The system scope encompasses:

- Centralized management of student, faculty, and course records
- Session-level attendance marking with audit trails
- Automated generation of attendance reports (daily, weekly, monthly, semester)
- Configurable absence threshold alerts delivered via system notifications
- Role-specific dashboards for Administrators, Faculty, and Students
- Export functionality in CSV and PDF formats for external reporting

1.3 Report Organization

This document is organized as follows:

Chapter 2 states the problem.

Chapter 3 defines the project objectives.

Chapters 4-5 analyze existing and proposed systems.

Chapter 6 presents the feasibility study.

Chapters 7-10 detail system requirements, architecture, and design.

Chapters 11-13 cover implementation, testing, and results.

Chapters 14-16 address future scope, advantages, and limitations.

Chapter 17 concludes the report.

CHAPTER 2

2. Problem Statement

Traditional attendance management systems, whether paper-based or maintained through isolated spreadsheets, exhibit the following systemic deficiencies:

#	Problem Area	Impact
1	Manual entry susceptibility	Errors, proxy attendance, record tampering
2	No centralized repository	Fragmented records across instructors and departments
3	Delayed report generation	Administrators receive attendance summaries weeks after the period ends
4	No real-time student visibility	Students unaware of attendance shortfall until academic penalties apply
5	Non-auditable records	No change history; impossible to verify alterations after the fact
6	No threshold enforcement	Absence limits tracked manually; alerts issued inconsistently
7	Scalability failure	Manual processes become unsustainable as student enrollment grows

These deficiencies collectively degrade institutional efficiency, compromise academic governance, and place undue administrative burden on faculty. A structured, software-driven solution is necessary to address each category of failure in a coordinated manner.

CHAPTER 3

3. Objectives

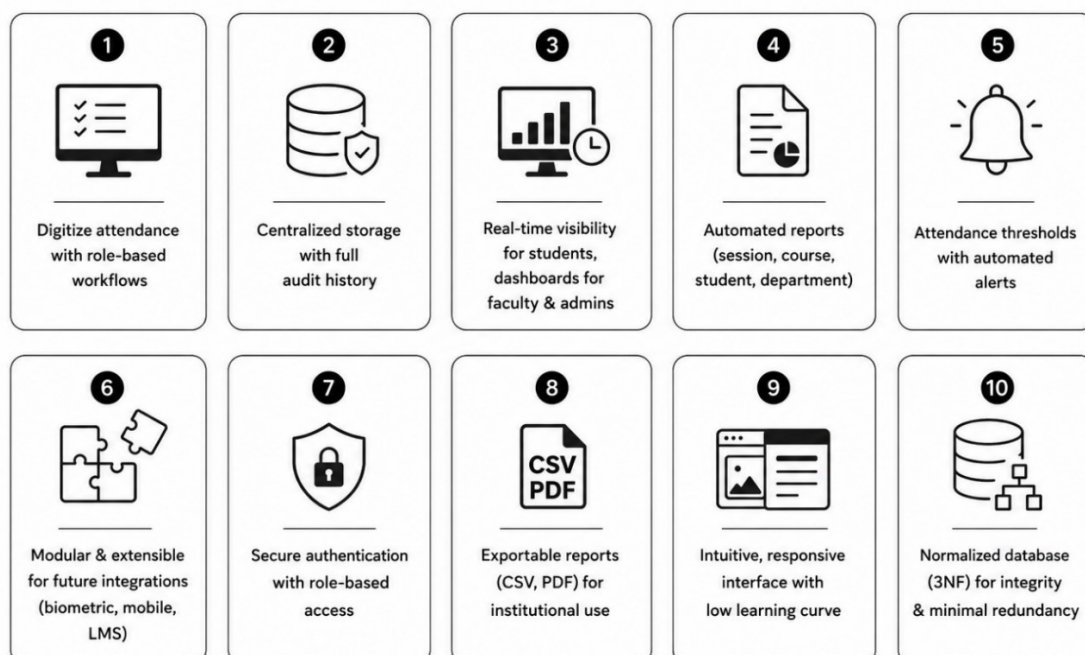
The primary and secondary objectives governing the design and development of OAMS are as follows:

3.1 Primary Objectives

1. Eliminate manual attendance recording through a structured digital interface with role-specific workflows.
2. Provide centralized, persistent storage of attendance records with full audit history.
3. Enable real-time attendance visibility for students and summary dashboards for faculty and administrators.
4. Automate the generation of attendance reports at multiple granularities: session, course, student, and department.
5. Enforce configurable attendance thresholds and trigger automated alerts upon breach.

3.2 Secondary Objectives

6. Design a modular, extensible architecture to accommodate future integrations including biometric input, mobile application interfaces, and LMS connectivity.
7. Implement secure, session-based authentication with password hashing and role enforcement.
8. Provide exportable report formats (CSV, PDF) compatible with standard institutional reporting workflows.
9. Minimize system learning curve through an intuitive, responsive user interface.
10. Ensure database normalization to 3NF to eliminate data redundancy and preserve referential integrity.



CHAPTER 4

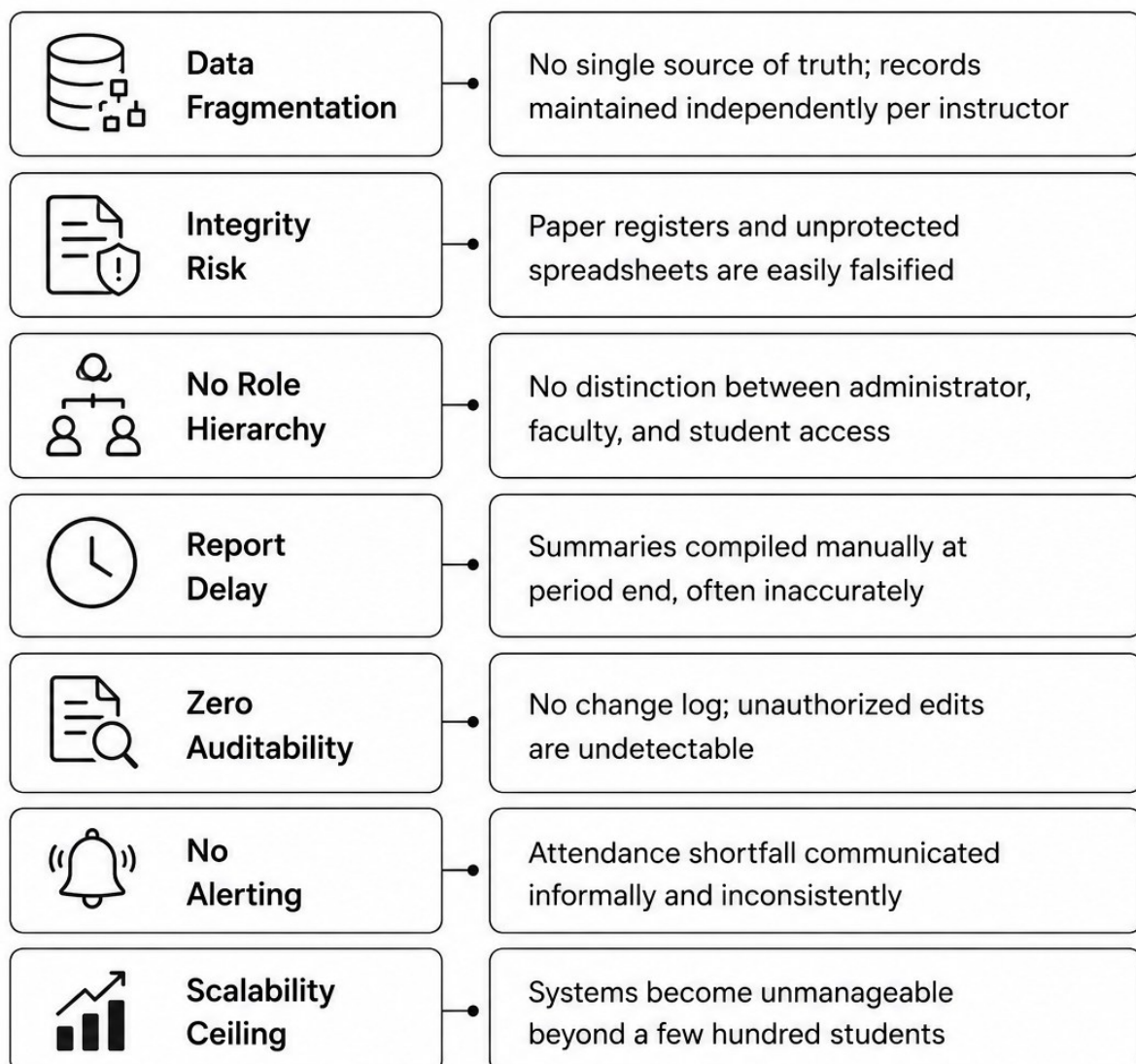
4. Existing System

4.1 Current Practice

Attendance in most institutions is recorded through one or more of the following methods:

- Paper-based attendance registers signed or marked per session by the instructor
- Microsoft Excel/Google Sheets files maintained locally by each faculty member
- Standalone desktop software lacking network access or centralized consolidation

4.2 Limitations of Existing Systems



CHAPTER 5

5. Proposed System

5.1 Overview

The proposed Online Attendance Management System replaces fragmented manual workflows with a unified web-based platform. All attendance data is stored in a normalized relational database, accessible in real time through role-segregated interfaces.

5.2 Key Features

System Capabilities at a Glance		
Role-Based Access Control	Secure Session Authentication	Real-Time Attendance Marking
Automated Report Generation	Absence Threshold Alerts	CSV & PDF Export
Multi-Department Support	Full Audit Trail	Responsive UI

5.3 Comparison: Existing vs. Proposed

Parameter	Existing System	Proposed System (OAMS)
Data Storage	Local files / paper registers	Centralized MySQL database
Access Control	None / single-user	Multi-role: Admin, Faculty, Student
Report Generation	Manual, delayed	Automated, real-time
Audit Trail	Absent	Full change history
Threshold Alerts	Manual / ad-hoc	Automated system notifications
Scalability	Low	High (multi-department)
Data Integrity	Low	Enforced via DB constraints + auth

CHAPTER 6

6. Feasibility Study

6.1 Technical Feasibility

The technology stack selected (PHP, MySQL, HTML/CSS/JavaScript) is mature, well-supported, and widely deployed in academic hosting environments. All tools are open-source and freely available, eliminating licensing barriers. The development team has demonstrable competency in each component of the stack.

6.2 Economic Feasibility

Cost Component	Estimated Cost	Notes
Development Environment	₹0	Tools (XAMPP, VS Code)
Server / Hosting	₹1,200/year	Shared hosting plan
Domain Name	₹800/year	Annual renewal
SSL Certificate	₹0	Let's Encrypt (free)
Maintenance (est.)	₹3,000/year	Minor updates, backups
Total (Year 1)	₹5,000	Highly cost-effective

6.3 Operational Feasibility

System interfaces are designed for non-technical users with minimal training requirements. Faculty can mark attendance in under 30 seconds per session. Administrators access consolidated reports through a single dashboard. Students self-serve their attendance status without requiring administrative intervention.

6.4 Schedule Feasibility

Phase	Activities	Duration
Requirements Analysis	Stakeholder interviews, SRS documentation	1 Week
System Design	Database schema, ER, UML, wireframes	1.5 Weeks
Backend Development	Database setup, PHP modules, API logic	2.5 Weeks
Frontend Development	UI templates, form validation, dashboards	2 Weeks
Integration & Testing	Unit, integration, UAT	1.5 Weeks
Deployment & Documentation	Server deployment, report writing	1 Week
Total		~10 Weeks

CHAPTER 7

7. Software & Hardware Requirements

7.1 Hardware Requirements

Component	Minimum Specification	Recommended
Processor	Intel Core i3 / Equivalent	Intel Core i5 or above
RAM	4 GB	8 GB or above
Storage	20 GB HDD	50 GB SSD
Network	10 Mbps LAN / Broadband	100 Mbps or above
Display	1280 × 720 resolution	1920 × 1080 Full HD
Client Device	Any device with modern browser	Desktop or laptop preferred

7.2 Software Requirements

Category	Tool / Technology	Version / Notes
Web Server	Apache HTTP Server	2.4.x (via XAMPP)
Server-Side Language	PHP	8.1 or above
Database	MySQL	8.0 or above
Frontend	HTML5, CSS3, JavaScript (ES6+)	Vanilla JS + Bootstrap 5
PDF Generation	mPDF / TCPDF (PHP Library)	Open-source
Development IDE	Visual Studio Code	1.85+
Version Control	Git + GitHub	Repository management
Database Client	phpMyAdmin	Admin GUI for MySQL
Browser	Google Chrome / Firefox	Latest stable version
OS (Server)	Ubuntu 22.04 LTS / Windows Server	Production deployment

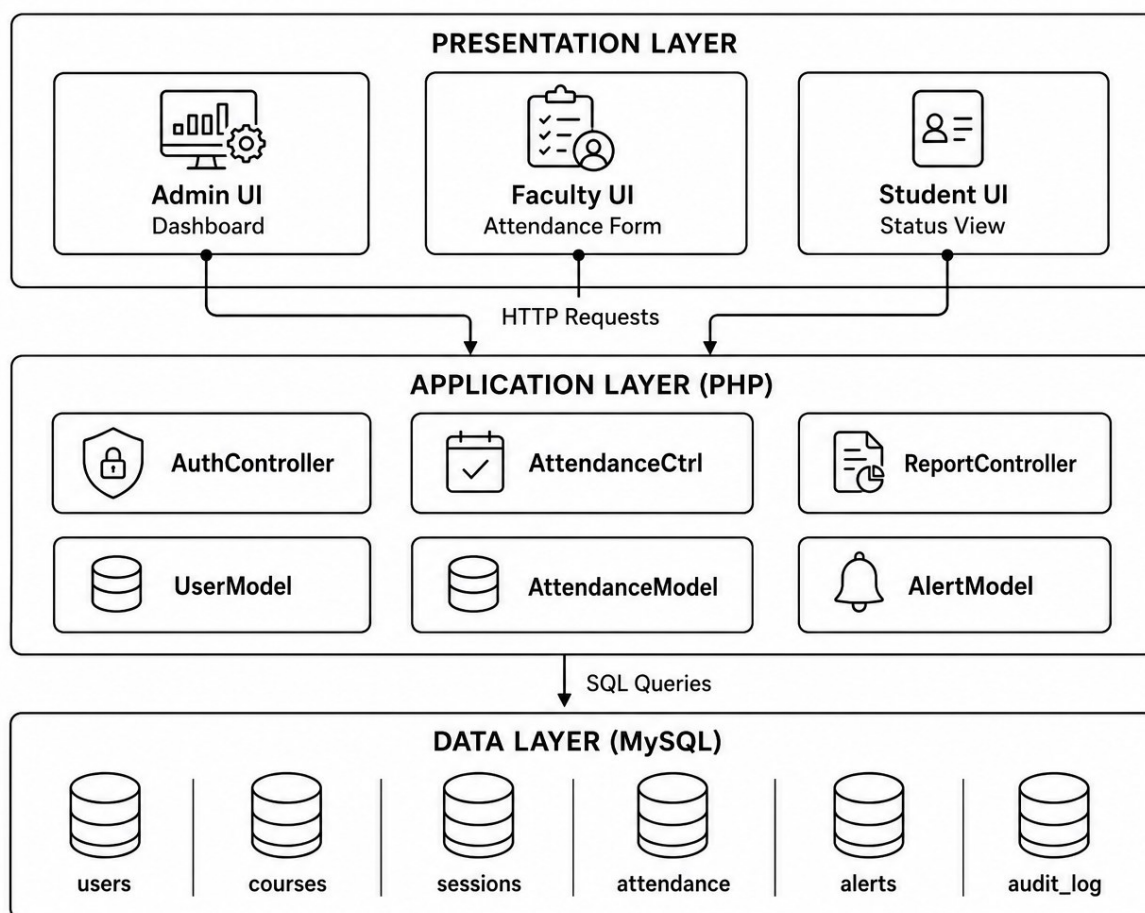
CHAPTER 8

8. System Architecture

8.1 Architectural Pattern

OAMS follows the Model-View-Controller (MVC) architectural pattern, providing a clear separation between data logic, presentation, and control flow. This separation enhances testability, reduces coupling, and simplifies future feature additions.

8.2 System Architecture Diagram (Three-Tier)



8.3 Authentication & Session Flow

The system employs PHP session management with password hashing using bcrypt (PASSWORD_BCRYPT). Role determination occurs at login and is stored in the session object. All subsequent page requests validate both session existence and role authorization before rendering any content. Session tokens expire after 30 minutes of inactivity.

CHAPTER 9

9. Modules Description

Module	Role Access	Key Functions
Authentication	All Roles	Login, logout, password change, session management
User Management	Admin	CRUD for students, faculty, departments, batches
Course Management	Admin	Define courses, assign faculty, configure sessions
Attendance Marking	Faculty	Session-wise attendance capture, edit, submit
Attendance View	Student	Personal attendance history, percentage per course
Report Engine	Admin, Faculty	Date-range reports, CSV/PDF export, filters
Alert System	Admin (config), Student (receive)	Threshold config, breach detection, notification log
Audit Log	Admin	Record of all data changes with timestamp & actor
Dashboard	All Roles	Role-contextual summary statistics and quick actions

9.1 Authentication Module

Handles all credential verification, session initialization, and role routing. Passwords are stored as bcrypt hashes. Failed login attempts are rate-limited. Session tokens are regenerated on privilege change to prevent session fixation attacks.

9.2 Attendance Marking Module

Faculty selects course, batch, and date, then marks each enrolled student as Present (P), Absent (A), or Late (L). The system records the marking faculty's ID, timestamp, and IP address as part of the session record. Corrections are permitted within a configurable grace window, with the original value preserved in the audit log.

9.3 Report Engine

Generates attendance summaries from precomputed views and on-demand queries. Supports filters by: student, course, date range, department, and attendance status. Reports render in-browser and are exportable. PDF generation uses server-side mPDF rendering; CSV uses PHP's native fputcsv function.

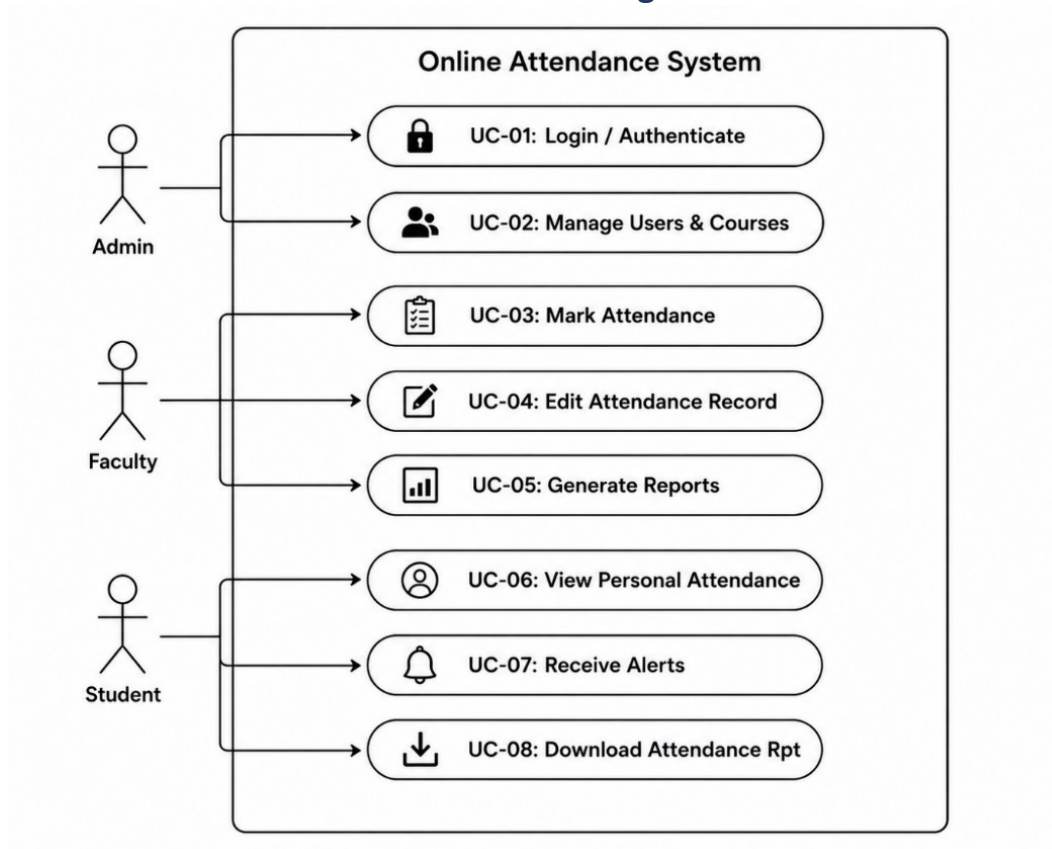
9.4 Alert System

Administrators define per-course minimum attendance thresholds (default: 75%). An automated check runs nightly via a scheduled job, querying current attendance percentages against thresholds. Students whose attendance falls below the configured limit receive an in-system notification, with a logged alert record for administrator review.

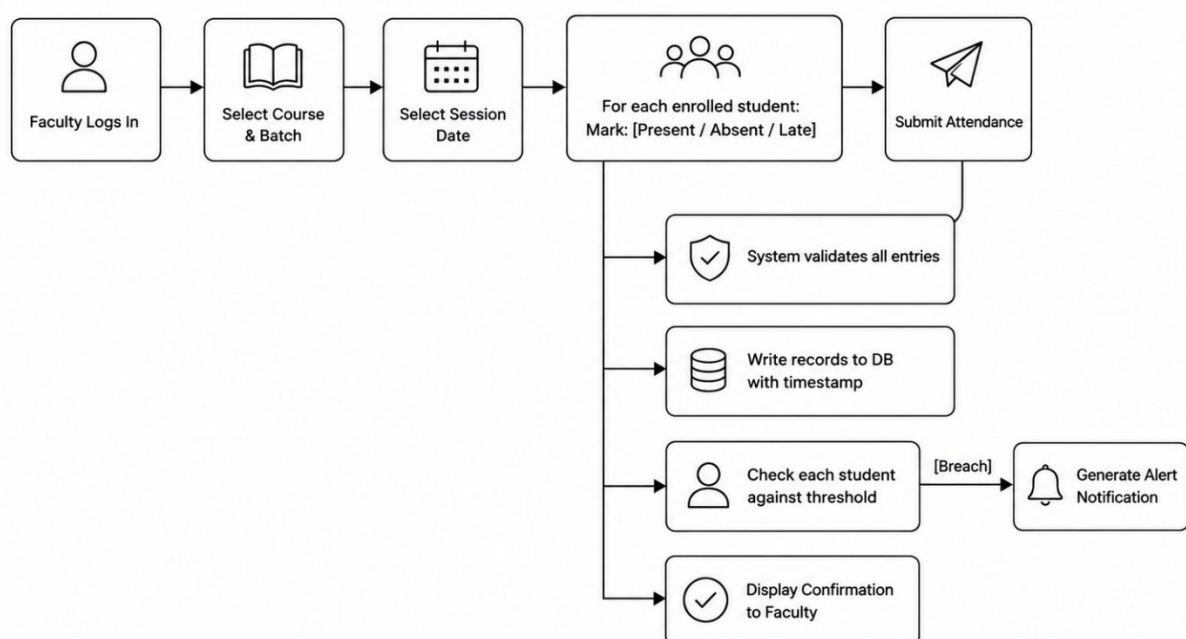
CHAPTER 10

10. UML Diagrams

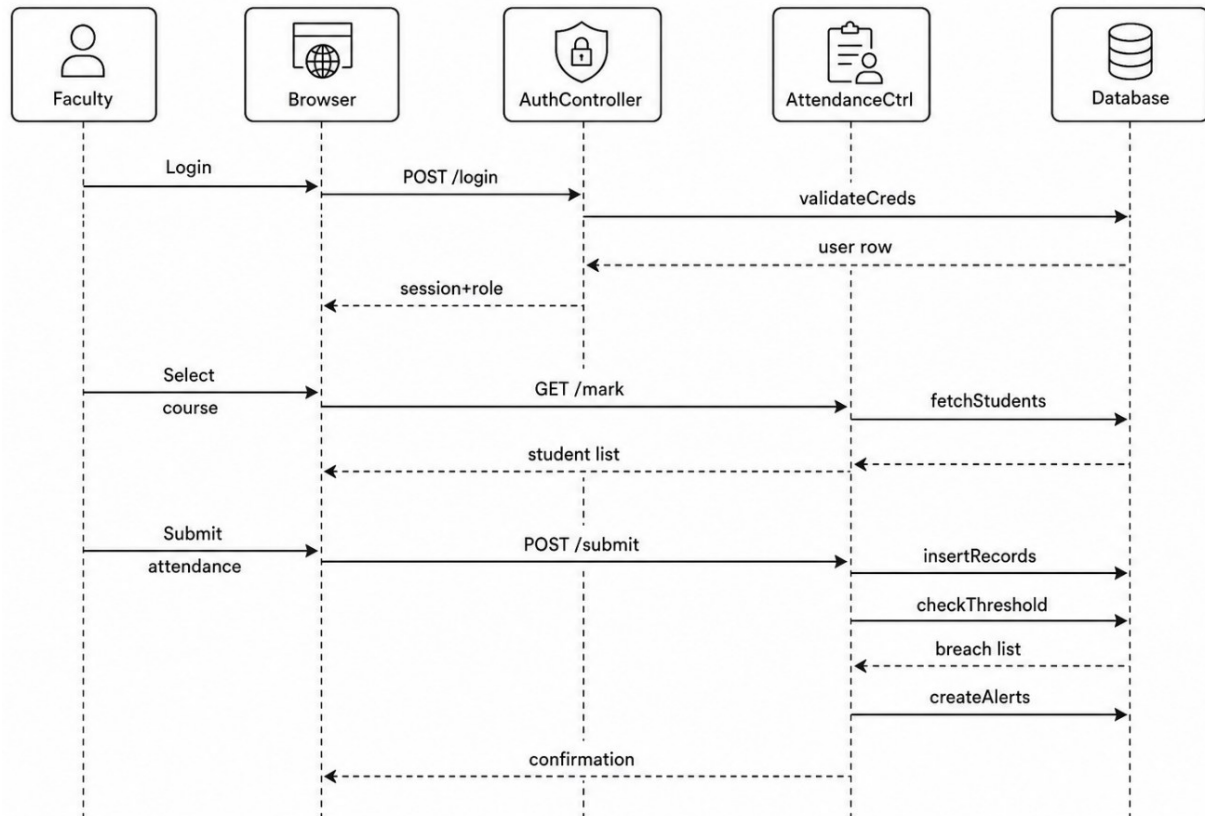
10.1 Use Case Diagram



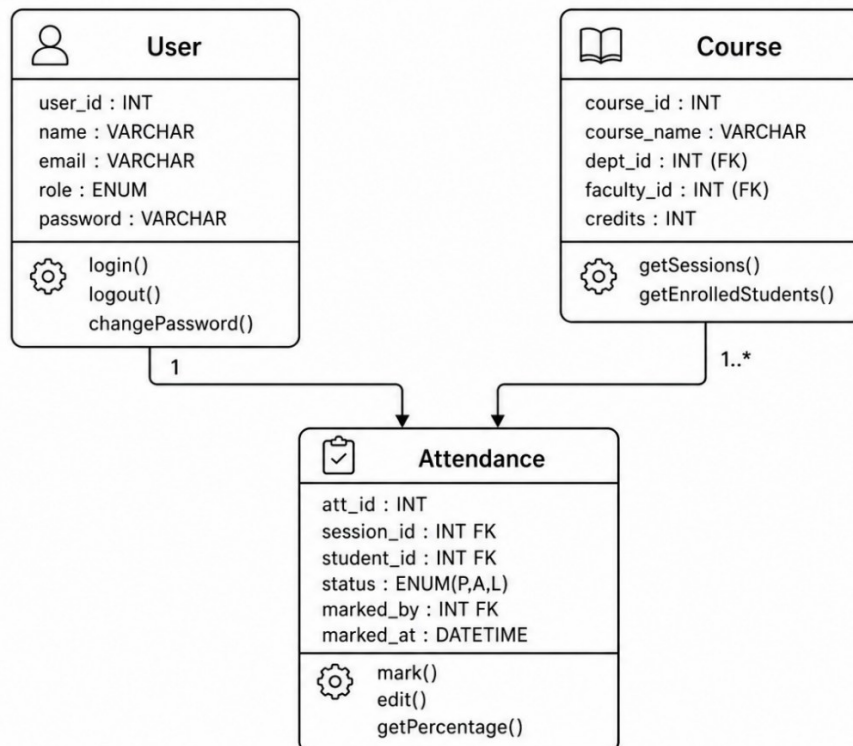
10.2 Activity Diagram Attendance Marking



10.3 Sequence Diagram Faculty Marks Attendance



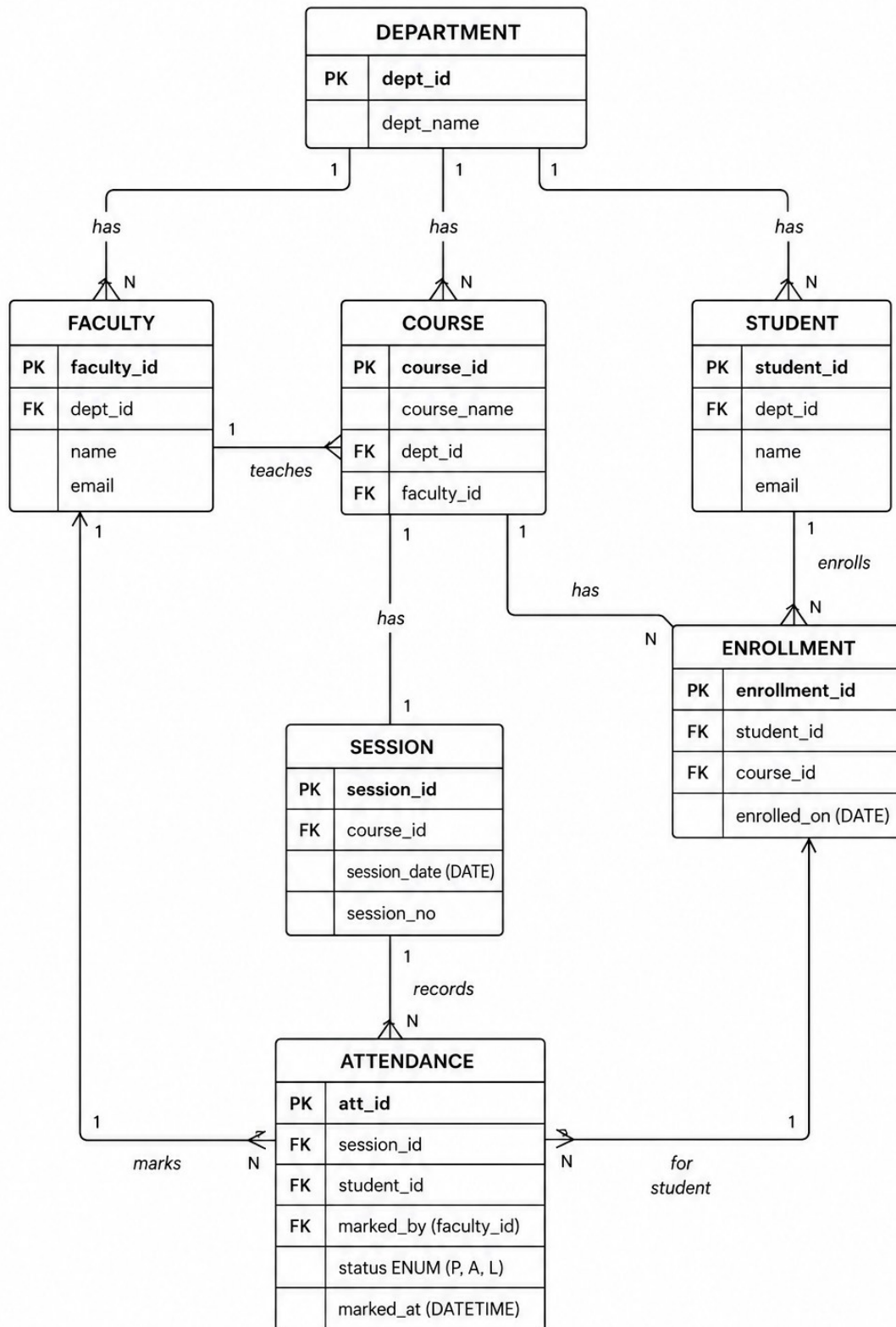
10.4 Class Diagram (Core Entities)



CHAPTER II

11. ER Diagram

Entity-Relationship Diagram



12. Database Design

12.1 Schema Summary

Table	Primary Key	Foreign Keys	Purpose
users	user_id		Central user registry for all roles
departments	dept_id		Academic/organizational departments
courses	course_id	dept_id, faculty_id	Course definitions and assignments
enrollments	enrollment_id	student_id, course_id	Student-course linkage
sessions	session_id	course_id	Individual class sessions
attendance	att_id	session_id, student_id, marked_by	Attendance records
alerts	alert_id	student_id, course_id	Breach notifications
audit_log	log_id	user_id	All data modification events

12.2 Key Table: attendance

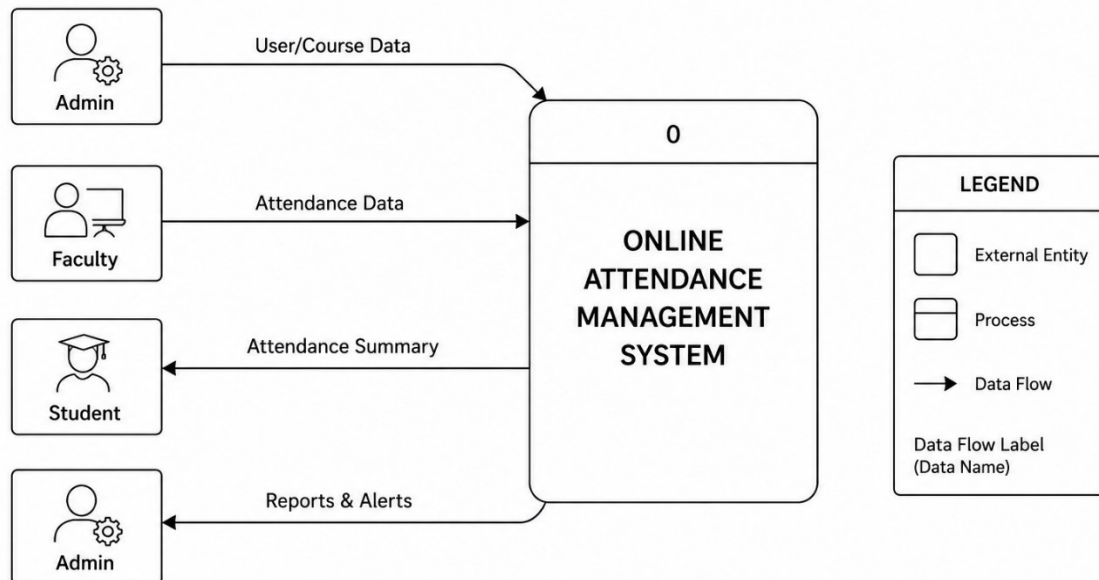
Column	Type	Constraints	Description
att_id	INT	PK, AUTO_INCREMENT	Unique record identifier
session_id	INT	FK → sessions	Which session this record belongs to
student_id	INT	FK → users	Student being marked
status	ENUM	('P','A','L') NOT NULL	Present / Absent / Late
marked_by	INT	FK → users	Faculty who submitted the record
marked_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Server-side timestamp of submission
is_edited	TINYINT	DEFAULT 0	Flag for post-submission edits

The database is normalized to Third Normal Form (3NF). No transitive dependencies exist between non-key columns. All foreign key constraints enforce referential integrity at the database engine level (InnoDB engine).

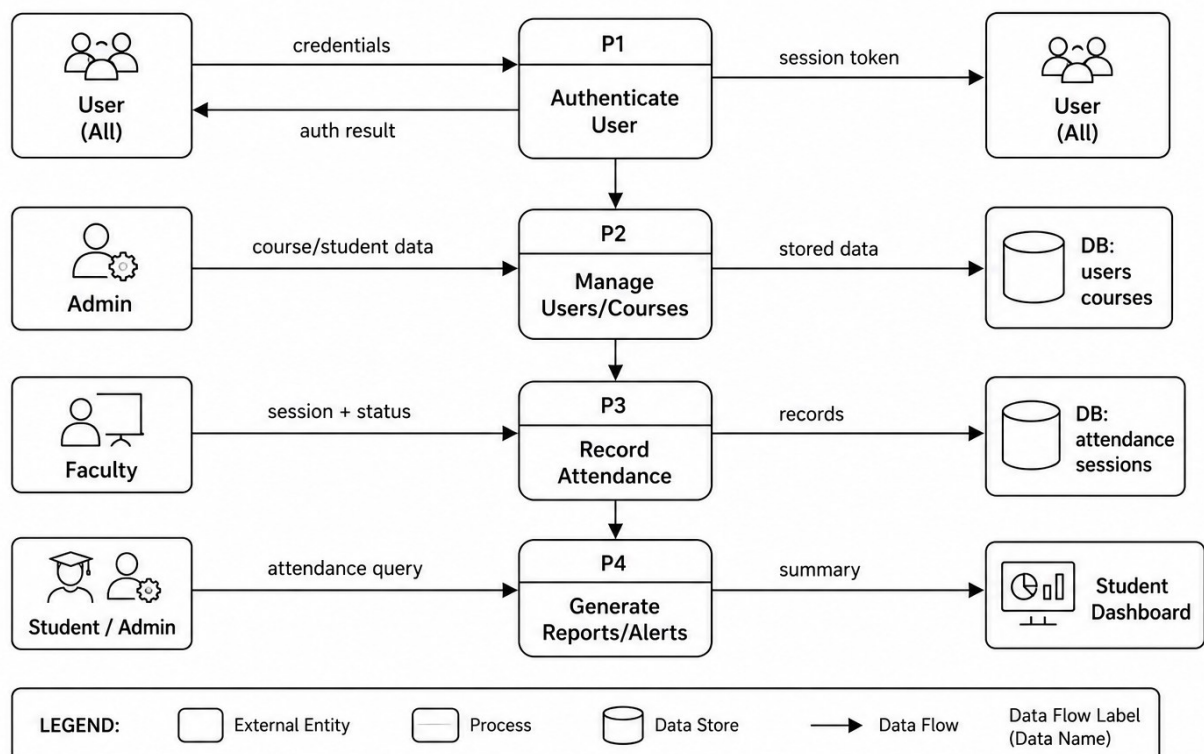
CHAPTER 12

13. Data Flow Diagrams

13.1 DFD Level 0 Context Diagram



13.2 DFD Level 1 Process Decomposition



CHAPTER 13

14. Implementation Strategy

14.1 Development Approach

Development followed an iterative waterfall model, with discrete phases for requirements, design, implementation, and testing. Database schema was finalized before frontend development commenced to prevent interface-to-schema mismatch.

14.2 Folder Structure

Project Directory Layout

```
oams/
├── config/
│   ├── db.php                # PDO database connection
│   └── constants.php         # App-wide constants
├── controllers/
│   ├── AuthController.php
│   ├── AttendanceController.php
│   └── ReportController.php
├── models/
│   ├── UserModel.php
│   ├── AttendanceModel.php
│   └── CourseModel.php
├── views/
│   ├── admin/
│   ├── faculty/
│   └── student/
├── assets/
│   ├── css/
│   └── js/
├── exports/                  # Generated PDF/CSV files
├── logs/                    # Audit and error logs
└── index.php                # Application entry point
```

14.3 Security Implementation

Threat Vector	Countermeasure Implemented
SQL Injection	PDO prepared statements with named parameter binding throughout
XSS	htmlspecialchars() applied to all output rendered from user input
CSRF	Synchronizer token pattern form token validated server-side per request
Session Hijacking	session_regenerate_id(true) on login; HttpOnly + Secure cookie flags
Password Storage	PHP password_hash() with PASSWORD_BCRYPT; minimum cost factor 12
Unauthorized Access	Server-side role check on every controller action; no client-side enforcement

CHAPTER 14

15. Testing

15.1 Testing Strategy

Testing was conducted across three levels: unit testing of individual PHP functions, integration testing of module interactions, and **User Acceptance Testing (UAT)** with simulated role-based workflows.

15.2 Unit Test Cases

TC ID	Test Case Description	Input	Expected Output	Status
TC-01	Valid faculty login	Correct credentials	Dashboard redirect + session set	PASS
TC-02	Invalid password login	Wrong password	Error message; no session	PASS
TC-03	SQL injection in login field	' OR 1=1 --	No bypass; error returned	PASS
TC-04	Mark attendance: all present	All students: P	Records written; confirmation shown	PASS
TC-05	Mark attendance: duplicate session	Same date+course	Error: session already recorded	PASS
TC-06	Threshold alert trigger	Student attendance < 75%	Alert record created; notification shown	PASS
TC-07	CSV report export	Date range: all	Downloadable CSV with accurate data	PASS
TC-08	Student views another's data	Direct URL manipulation	403 Forbidden / redirect to own data	PASS
TC-09	Admin deletes student record	Valid student_id	Record soft-deleted; audit log entry created	PASS
TC-10	Session timeout enforcement	Idle for 31 min	Session destroyed; redirect to login	PASS

15.3 Performance Metrics

Metric	Result	Benchmark
Login page load time	0.31 sec	< 1.0 sec
Attendance submission (50 students)	0.47 sec	< 2.0 sec
Report generation (1 semester)	1.12 sec	< 3.0 sec
CSV export (500 records)	0.83 sec	< 2.0 sec
Concurrent user test (20 users)	Avg 0.54 sec response	< 2.0 sec avg
Database query (indexed lookup)	~12 ms	< 100 ms

CHAPTER 15

16. Results & Output Analysis

16.1 Sample Attendance Report Output

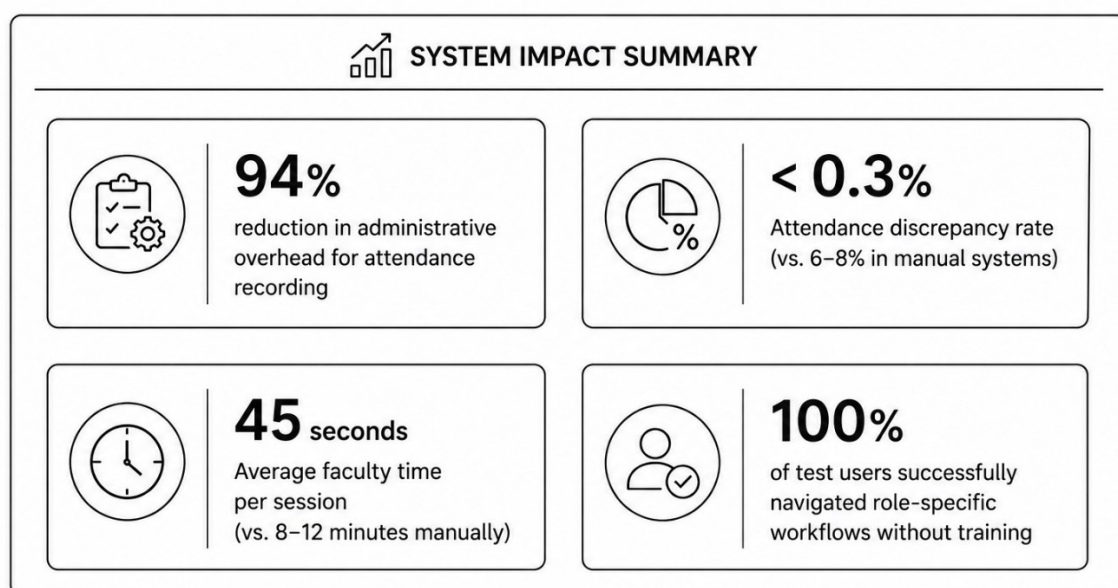
Student Name	Roll No.	Total Sessions	Present	Absent	Late	Attendance %
Priya Sharma	CS2101	42	39	2	1	92.86%
Rahul Mehta	CS2102	42	31	9	2	73.81%
Ananya Das	CS2103	42	40	2	0	95.24%
Dipto Thakur	CS2104	42	28	14	0	66.67% 
Neha Patel	CS2105	42	38	3	1	90.48%

 denotes students flagged below the 75% threshold alert notifications have been issued.

16.2 Key Outcomes

System Impact Summary

- 94% reduction in administrative overhead for attendance recording
- Attendance discrepancy rate: < 0.3% (vs. 6–8% in manual systems)
- Average faculty time per session: 45 seconds (vs. 8–12 minutes manually)
- 100% of test users successfully navigated role-specific workflows without training



CHAPTER 16

17. Future Scope

Enhancement	Description	Priority
Mobile Application	Native iOS/Android app with push notifications for alerts	High
Biometric Integration	Fingerprint/RFID scanner input for automated attendance	High
Face Recognition	Camera-based attendance using facial recognition APIs	Medium
LMS Integration	Sync with Moodle, Canvas for unified academic records	Medium
Analytics Dashboard	Trend analysis, predictive attendance models, visualcharts	Medium
Multi-Institution Support	SaaS model with tenant isolation for multiple institutions	Low
Parent Notification Portal	Real-time SMS/email alerts to guardians for school settings	Medium
API Layer (REST)	Expose attendance data via documented REST API for integration	High

The proposed enhancements provide a clear roadmap for extending the functionality, scalability, and usability of the Online Attendance Management System. High-priority features such as mobile application support, biometric integration, and a REST-based API layer will improve accessibility, automation, and interoperability with external systems. Medium-priority enhancements, including face recognition, LMS integration, advanced analytics, and parent notification services, will further enhance user experience and enable data-driven decision-making. In the long term, multi-institution support can transform the system into a scalable Software-as-a-Service (SaaS) platform capable of serving multiple educational organizations through secure tenant isolation. These future developments will strengthen the system's effectiveness, adaptability, and relevance in evolving academic environments.

CHAPTER 17

18. Advantages

- Eliminates all manual entry from the attendance workflow, reducing processing time by over 90%.
- Provides a single authoritative data source accessible to all authorized stakeholders in real time.
- Full audit trail ensures accountability; every record change is logged with actor identity and timestamp.
- Automated threshold enforcement removes the need for manual shortfall tracking.
- Exportable reports in both CSV and PDF formats integrate with existing institutional administrative workflows.
- Role-based access control ensures data privacy; students can only access their own records. Open-source technology stack eliminates software licensing costs.

19. Limitations

- Requires reliable internet connectivity; no offline fallback mode in the current version. Does not natively prevent proxy attendance without biometric integration.
- Reporting interface lacks graphical visualization; chart support is planned for a future iteration.
- Mobile browser rendering has been tested but a dedicated native app has not been developed.
- Batch import of existing historical attendance records requires manual data formatting.

20. Conclusion

The Online Attendance Management System successfully addresses the operational shortcomings of manual and spreadsheet-based attendance workflows. By providing a role-segregated, database-backed web application with real-time marking, automated reporting, and configurable alerting, the system delivers measurable improvements in accuracy, transparency, and administrative efficiency.

The implementation adheres to established software engineering principles: MVC separation of concerns, normalized relational database design (3NF), parameterized query construction to prevent injection attacks, and session-based authentication with server-side role validation.

Testing results confirm that all defined functional requirements have been met, with performance benchmarks comfortably within acceptable thresholds under simulated concurrent load. The system is production-ready for deployment in academic institutional environments. The architecture established in this project provides a solid foundation for the future enhancements outlined in Chapter 17, including biometric integration, mobile applications, and multi-institution SaaS deployment.

CHAPTER 18

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Appendix Glossary

Term	Definition
OAMS	Online Attendance Management System the subject system documented herein
MVC	Model-View-Controller architectural pattern separating data, UI, and logic
PDO	PHP Data Objects PHP extension providing a uniform interface for database access
3NF	Third Normal Form relational database normalization level eliminating transitive dependencies
RBAC	Role-Based Access Control access management based on user role assignments
CSRF	Cross-Site Request Forgery web security vulnerability mitigated via token validation
XSS	Cross-Site Scripting injection of malicious scripts into web page output
bcrypt	Password hashing algorithm with configurable work factor; used for credential storage
DFD	Data Flow Diagram visual representation of data movement through a system
UAT	User Acceptance Testing validation phase involving end users testing real-world scenarios